Look-Ahead Handoffs and Runtime-Adaptive Bank Eligibility for a Tiled Neural-Network Scratchpad

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Abstract

Scratchpad buffering and controller handoff timing can independently limit a neural-network accelerator's multiply-accumulate (MAC) utilization. We implement runtime bank eligibility and completion-aware look-ahead handoffs in one Verilog design with four physical RAM banks, one loader, and one MAC. The depth policy uses a 32-cycle starvation window and integer 8/2 hysteresis thresholds to permit new allocations to two, three, or four banks while allowing existing tiles to drain. The handoff scheduler accepts a new load or MAC descriptor on the edge that the previous operation finishes; per-tile transfer and compute service times remain unchanged. Self-checking RTL simulations validate arithmetic results, bank ownership, and per-tile service intervals under stable, variable-latency, and phase-changing workloads. For 64 balanced tiles, adaptive look-ahead execution requires 522 cycles compared with 713 for fixed four-bank allocation with registered handoffs, a 26.79% reduction. Variable and phase-changing workloads take 925 versus 983 and 1808 versus 1991 cycles, reductions of 5.90% and 9.19%. The corresponding average allocation limits are 2.061, 3.069, and 2.619 banks. The proposed controller combines handoff scheduling and eligibility control while retaining the same four physical RAM banks, one loader, and one MAC. The measurements characterize execution cycles and allocation eligibility; clock frequency, SRAM power, and physical area are not measured. **Index Terms**— neural-network accelerator, scratchpad, bank allocation, look-ahead scheduling, Verilog RTL, adaptive control.

1 Introduction

Data movement and reuse often dominate accelerator design decisions [1]. A scratchpad keeps a tile near the MAC, and ping-pong buffering can load tile $t + 1$ while tile t is computed. Two banks, however, hold only two independent tiles. A third or fourth bank can help when the transfer stream is irregular; when computation is consistently slower, those extra eligible slots may provide no cycle benefit. A fixed four-bank policy cannot distinguish these regimes.

Buffer capacity alone does not determine utilization. Registered control interfaces can introduce idle cycles between successive operations even when a tile is available. This study evaluates bank eligibility and handoff timing as independent control dimensions within a four-bank scratchpad. Runtime eligibility restricts load allocations to two, three, or four banks, while completion-aware handoffs permit a resource to accept its next descriptor on the edge that its current operation finishes. Tile values, RAM timing, input order, and the single-loader, single-MAC organization are held constant.

The contributions are: (1) a runtime bank-allocation rule with explicit ownership invariants and safe draining during depth reduction; (2) loader and MAC handoff logic that eliminates control gaps without shortening per-tile service intervals; and (3) a controlled RTL evaluation against single buffering, ping-pong, and fixed four-bank allocation. The primary baseline, Fixed 4, uses four eligible banks and registered resource handoffs. The Proposed controller combines adaptive two/three/four-bank eligibility with completion-aware handoffs. Both use the same RAM and MAC organization. RH and LH denote registered and look-ahead handoffs when describing their control timing.

2 Background and Related Work

The survey by Sze *et al.* relates arithmetic reuse to movement among the memory hierarchy levels [1]. Eyeriss demonstrates distributed local storage and row-stationary data reuse [2]. The datacenter TPU uses a software-managed on-chip memory in a complete inference system [3]. Eyeriss v2 adds flexibility for layer shapes and sparsity [4]. These systems justify careful treatment of on-chip storage, but their frame rates and full-system speedups cannot be compared directly with our single-MAC tile-cycle model.

At the controller level, single buffering serializes transfer and compute. Ping-pong permits overlap but can block the loader while one bank is computing and the

other is occupied. A fixed deeper buffer offers more tile slots. Runtime eligibility adjusts the allocation limit in response to compute starvation, whereas look-ahead scheduling reduces the control latency between resource assignments. The static cyclic policy provides a fixed tile-to-bank mapping against which availability arbitration can be evaluated at equal bank count. Table 1 summarizes the scope and evaluation metrics of the related accelerator designs.

3 Architecture and RTL Implementation

3.1 Common Data Path and Bank Protocol

Figure 1 shows the implemented structure. Each tile contains T activation-weight pairs of width W per operand. A single loader writes one pair per active clock into a synchronous RAM bank. A read mux feeds one MAC, which reads one pair per clock and accumulates T products. The transfer target specifies the effective tile-load interval; the compute target may include latency after useful MAC operations. Four physical banks remain present in the runtime design. Bank-depth policies change the choice of a bank for the next tile. Handoff scheduling determines when the loader and MAC can accept their next descriptors.

The per-bank protocol in Fig. 2 has states EMPTY, LOADING, READY, and COMPUTING. The loader may claim only an EMPTY bank. Finishing the programmed load interval makes that bank READY. The MAC claims a READY bank and releases it to EMPTY only after the result is registered. Each bank retains a tile ID, so availability dispatch need not return results in input order. Under RH control, a freed bank becomes eligible for arbitration on the following clock. LH control uses completion predicates to permit bank release and descriptor acceptance on the same edge.

Table 2 isolates both bank count and scheduling policy. Single and ping-pong assign a tile to a fixed bank by its ID and dispatch strictly in order. The four-bank availability policy searches EMPTY banks round-robin for loading and READY banks round-robin for computing. The static cyclic proxy has the same four tile slots but a fixed ID-to-bank mapping, so it can test whether availability arbitration adds value beyond capacity. The runtime policy reuses the availability arbiter, but rejects an EMPTY bank with index at or above the current allocation limit.

3.2 Runtime Bank-Limit Algorithm

The proposed limit $B_c \in \{2, 3, 4\}$ starts at two and applies to new load allocations. The compute arbiter always searches all four banks. Thus, if the limit falls from four to three while bank 3 is READY or COMPUTING, its tile remains valid and can finish. No migration, invalidation, or discarded partial sum is required. This asymmetry between allocation and dispatch is the safety condition for a downward transition.

At each measured clock, a narrow starvation event s_c is one when work remains, the MAC is free, no dispatch is pending, and no bank is READY. The controller accumulates these events over $L = 32$ clocks. The update at the window boundary uses the just-observed event as well as the stored count. Figure 3 and the pseudocode in Fig. 4 state the entire rule. High and low thresholds are $H = 8$ and $D = 2$. Counts from three through seven hold the current mode; limits saturate at two and four. Only comparisons and a one-step increment or decrement are required in RTL.

3.3 Completion-Aware Look-Ahead Handoffs

The RH MAC interface registers a start request and observes a registered completion, leaving three idle clocks between successive MAC jobs even when data are queued. The loader similarly leaves one clock between transfers when another bank is free. A depth restriction alone cannot remove these costs or add scheduling opportunities beyond fixed four. The proposed controller removes these gaps by enabling completion-aware issue in both the loader and the MAC.

With loader look-ahead, the scheduler accepts the next descriptor when the current transfer reaches its unchanged target, and may reclaim a bank whose MAC operation completes on that edge. With MAC look-ahead, the next descriptor is chosen combinationally when the engine is idle or finishing. Eligible sources are a READY bank or the bank whose load finishes on this edge. Let L_f and M_f denote these completion predicates. Then the effective availability conditions are

$$E_b^* = (\text{state}_b = \text{EMPTY}) \vee (M_f \wedge b = b_{\text{MAC}}), \quad (1)$$

$$R_b^* = (\text{state}_b = \text{READY}) \vee (L_f \wedge b = b_{\text{load}}). \quad (2)$$

The allocation-limit check still applies to E_b^*. The MAC searches all banks using R_b^*, preserving safe drain on a limit decrease.

The MAC asserts M_f only after all T synchronous RAM responses have been accumulated and its programmed service target has elapsed. On a chained acceptance edge, nonblocking assignments capture the completed result and tile ID before resetting the accumulator for the next job. A completing load can feed a new MAC job safely: the last write occurs on the acceptance edge, while the first read of the new job occurs on the following edge. Bank release uses M_f, independently of the delayed result notification, so the notification cannot accidentally free a newly assigned bank.

Table 1: Storage and control scope of related accelerator designs. Published performance values refer to their respective workloads and platforms; they do not form a common numerical comparison.

Work	Main storage/control scope	Reported evidence	Relation to this paper
Eyeriss [2]	Distributed local storage and row-stationary reuse in a CNN chip	AlexNet convolution, batch 4: 35 frames/s and 0.0029 DRAM accesses/MAC	Motivates local reuse; different hardware and metric
TPU [3]	Software-managed on-chip memory feeding an inference systolic array	Production inference: roughly 15–30× over contemporary CPU/GPU systems	Full-system context only
Eyeriss v2 [4]	Flexible hierarchy for varied and sparse DNN layers	Sparse MobileNet, batch 1: 1470.6 inferences/s and 12.6× original Eyeriss	Flexibility context only
This work	Four-bank tile scratchpad with runtime depth and look-ahead handoffs	Matched RTL cycles, service traces, and average eligibility	Direct comparison with internal baselines

Table 2: Controlled policy distinction. All policies use the same RAM, operands, loader, MAC, and tile stream. "Limit" affects only new load allocations.

Policy	Banks	Load allocation	Compute dispatch	Limit
Single	1	Fixed bank; wait until EMPTY	Strict tile order	Fixed 1
Ping-pong	2	Tile ID modulo 2	Strict tile order	Fixed 2
Availability	4	Any EMPTY bank, round-robin	Any READY bank, round-robin	Fixed 4
Static cyclic proxy	4	Tile ID modulo 4	Strict tile order	Fixed 4
Runtime (proposed)	4	EMPTY bank below limit, round-robin	Any READY bank, round-robin	2, 3, or 4

3.4 Cycle Accounting and Metrics

For total measured cycles C, active MAC cycles C_{MAC}, completed tiles N, and effective limit B_c at clock c, we report

$$U = \frac{C_{\mathrm{MAC}}}{C}, \quad R = \frac{N}{C}, \quad \bar{B} = \frac{1}{C}\sum_{c=1}^{C} B_c. \tag{3}$$

The broad *compute-idle* or *stall* count is $C - C_{\mathrm{MAC}}$; it includes initial fill and handoff, whereas s_c is the narrower signal that drives adaptation. Speedup equals baseline cycles divided by candidate cycles. Average limit $\bar{B}$ measures allocation eligibility, not the number of physically disabled RAMs; no clock gating or bank power shutoff is modeled.

3.5 Control Invariants and Implementation Cost

With look-ahead disabled, the bank protocol gives two local safety conditions at every clock c:

$$\mathrm{allocate}(b, c) \Rightarrow (\mathrm{state}_b = \mathrm{EMPTY}) \wedge (b < B_c), \tag{4}$$

$$\mathrm{dispatch}(b, c) \Rightarrow \mathrm{state}_b = \mathrm{READY}. \tag{5}$$

The dispatch condition intentionally has no $b < B_c$ term. It permits a tile already in a high-index bank to drain after a limit decrease. The RTL also keeps at most one LOADING and one COMPUTING bank because each resource has one busy flag and one active-bank register. Result tile IDs let the testbench check exactly-once completion independently of dispatch order. With look-ahead enabled, E_b^* and R_b^* replace the state predicates; the completion edge transfers ownership without overlapping active users.

For four banks, the adaptation state consists of a window counter, a starvation counter, and the limit register; mode-time and transition counters are measurement instrumentation. The combinational eligibility search examines the bounded bank-state array, and the update uses integer comparisons rather than a division. This describes RTL structure, not post-synthesis area or critical-path timing.

3.6 Logic Synthesis

Yosys 0.69 is used to synthesize the parameterized RTL into generic gate netlists. Table 3 compares configurations with four four-word banks, eight-bit operands, 64 tiles, and fixed 8/8 service targets. The proposed controller requires 4713 logic/register cells compared with 3477 for Fixed 4, an increase of 35.55%. Both configurations pass the structural checks without inferred latches. Cell totals exclude five scope metadata records per netlist and include the measurement counters and flip-flop RAM mapping.

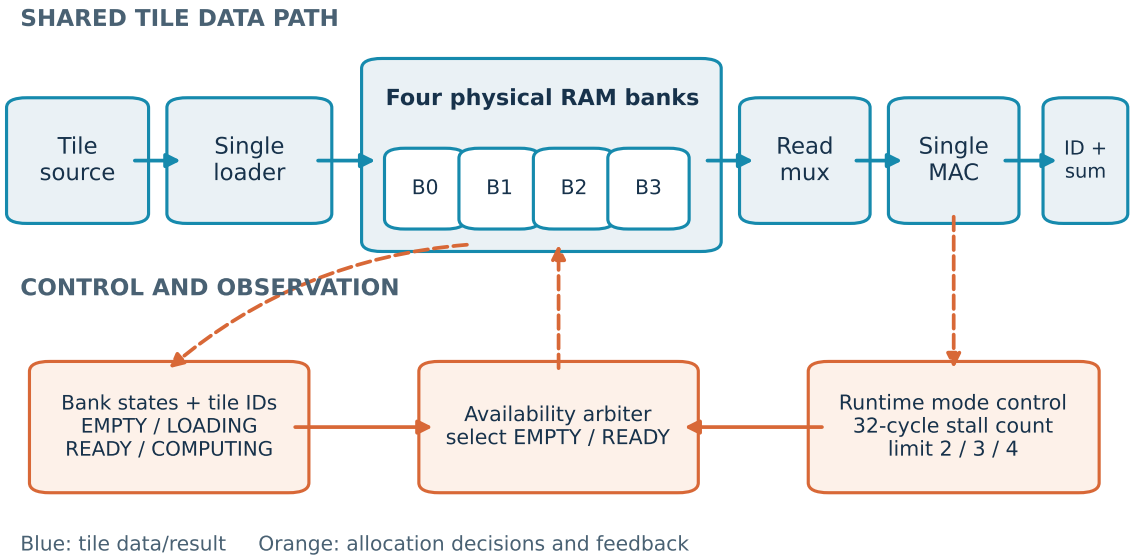


Figure 1: Implemented data path and runtime control. One loader and one MAC are shared across four physical RAM banks. The mode controller changes only which EMPTY banks may receive new tiles; the arbiter may still dispatch a READY tile from any bank. Blue arrows carry data; orange arrows carry control or feedback.

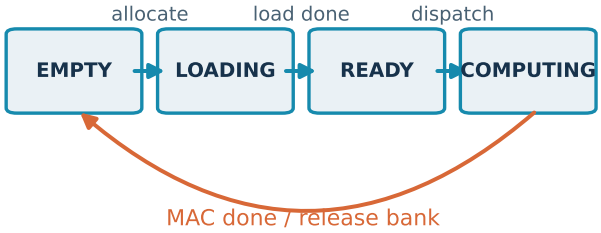


Figure 2: Logical per-bank state transitions. Look-ahead scheduling can combine completion and the next acceptance at one edge; an unfinished tile is never overwritten.

Table 3: Generic synthesis results under matched hardware and workload parameters. Register bits are included in the cell totals.

Configuration	Logic/register cells	Register bits
Fixed 4	3477	595
Proposed	4713	864

Figure 6 shows the synthesized RAM structure for a two-word, one-bit instance of the same parameterized RTL. Two enabled storage elements feed a read multiplexer and a registered output. This reduced instance exposes the storage and read-path logic; the accelerator measurements use the wider, deeper banks specified above.

Generic gate counts characterize circuit structure. Physical area and critical-path delay require technology mapping, SRAM macro selection, and timing analysis.

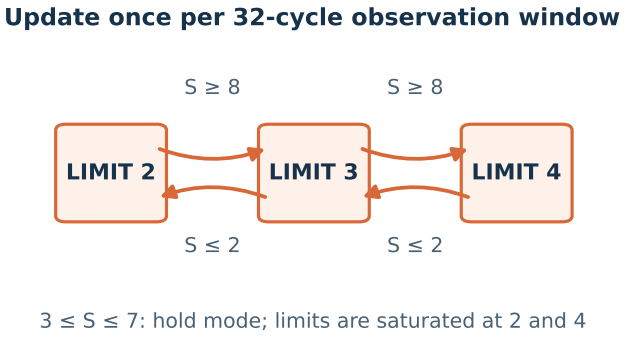


Figure 3: Hysteresis state machine. S is the accumulated no-READY count over one observation window; each transition changes the allocation limit by exactly one bank.

4 Verification and Experimental Setup

4.1 RTL Workload and Correctness Checks

The RAM, MAC, and controller are Verilog-2001-style synthesizable RTL. Only the testbench and experiment driver are simulation infrastructure. Icarus Verilog 14.0 compiles each parameterization with `-g2005`. For tile t and element i, the deterministic input operands are

$$a_{t,i} = ((t + i) \bmod 15) + 1, \tag{6}$$

$$w_{t,i} = ((3t + 2i) \bmod 15) + 1. \tag{7}$$

Algorithm: Runtime allocation-limit update

State: limit $B \leftarrow 2$, window count $q \leftarrow 0$, stall count $S \leftarrow 0$.
Each measured clock:

Find an EMPTY load bank with index $< B$; dispatch a READY bank from indices 0–3.

$s \leftarrow 1$ iff work remains, MAC is free, no start or completion is pending, and READY count is zero; else $s \leftarrow 0$.

$q \leftarrow q + 1$; $S \leftarrow S + s$.

if $q = 32$ **then**

  if $S \geq 8$ **then** $B \leftarrow \min(4, B + 1)$;

  else if $S \leq 2$ **then** $B \leftarrow \max(2, B - 1)$;

  $q \leftarrow 0$; $S \leftarrow 0$.

Figure 4: Pseudocode of the window update and bank eligibility. The RTL implements this with nonblocking sequential state updates; the count includes the event on the closing clock.

Table 4: Simulation and trace-based verification results.

Check	Result
Self-checking functional parameterizations	14/14 pass
RH configurations with checked tile results	129/129 pass
Independent cycle/operation checks	129/129 pass
VCD mode transitions checked	6
RH reference agreement: 13 metrics per case	129/129
Handoff correctness stress configurations	84/84 pass

The expected output is $y_t = \sum_{i=0}^{T-1} a_{t,i} w_{t,i}$. The testbench checks both y_t and tile ID at every completion, detects duplicates, and asserts that the one loader and one MAC never occupy multiple banks. It also checks legal bank allocation, bank-state accounting, and runtime mode limits. The functional suite contains 14 parameterizations. The RH matrix contains 129 configurations for which independent analysis verifies active cycle sums, operation counts, and structural throughput bounds. Thirteen reported metrics per configuration also agree with a frozen RH reference. A phase-case VCD establishes the six observed bank-limit transitions.

Handoff correctness is checked across 84 stress configurations covering single-bank reuse, non-power-of-two bank counts, minimum service targets, one-tile workloads, larger tiles, and variable or phase-changing timing. Issue scoreboards detect duplicate MAC acceptance, and bank assertions reject ownership and read/write collisions. Acceptance-to-completion traces verify each load and MAC service interval independently of the utilization counters. All reported configurations pass their arithmetic, protocol, and accounting checks.

4.2 Matched Experimental Matrix

All policies receive identical tile values and the same deterministic latency sequence in a matched scenario. There is one loader, one MAC, eight-bit operands, and a four-pair tile in the principal comparisons. Each bank has four addressable words unless stated otherwise. The transfer target has a floor of T clocks because the loader writes T pairs. The compute target has a floor of $T + 2$ clocks for the synchronous read pipeline. No clock frequency, DRAM channel timing, or physical SRAM macro is assumed.

Table 5: Experiment families and base transfer/compute targets in clocks per tile. Tiles contain four operand pairs except in the larger-tile family, which uses 16. Profiles are identical across policies.

Family	Tiles	Base	Profile
Fixed balanced	32	8/8	Constant
Fixed slower compute	64	8/20	Constant
Balanced adaptation	64	8/8	Constant
Fixed slower memory	64	32/8	Constant
Variable adaptation	64	8/8	Per-tile increments
Three-phase adaptation	96	8/8	Slow compute / irregular memory / slow compute
Minimum targets	64	4/6	Constant
Larger tile	16	16/20	Per-tile increments
One tile	1	8/8	Constant
Bank scaling	32	8/8	Variable; 2, 3, 4, 6, 8 banks

Table 5 summarizes the evaluated workload families. The broader matrix also varies tile count, tile size, and memory-to-compute ratio. The standard variable transfer increments repeat $[0, 8, 0, 7, 18, 0, 4, 13, 2, 10]$ by tile ID; compute increments repeat $[0, 4, 0, 8, 2]$. In the 96-tile phase experiment, the outer thirds add 12 compute clocks per tile, while the middle third uses base compute latency and transfer increments $[0, 16, 0, 10, 22, 1, 4, 19, 0, 12]$. These are deterministic stress profiles, not recorded neural-network traces. The exact parameters and raw logs are available in the project archive.

The primary comparison evaluates Fixed 4 and the proposed controller across eight workload families with 1, 16, 64, or 96 tiles. Each matched pair receives identical operands, service targets, and latency profiles. Single-buffer and ping-pong baselines establish the benefit of overlapping loading and computation. The 32-tile buffering and bank-count sweeps use RH control to characterize storage capacity and allocation policy.

5 Experimental Results

5.1 End-to-End Performance

Table 6 compares the proposed controller with Fixed 4. For the balanced 64-tile workload, the proposed design requires 522 cycles compared with 713 cycles for Fixed 4, a reduction of 26.79%. Both configurations perform 512 MAC-active and 512 load-active clocks; MAC-idle time decreases from 201 to ten clocks. The first tile finishes loading at edge 9, the last MAC finishes at edge 521, and its result is retired at edge 522. The active-cycle counts and completion trace attribute the reduction to handoff

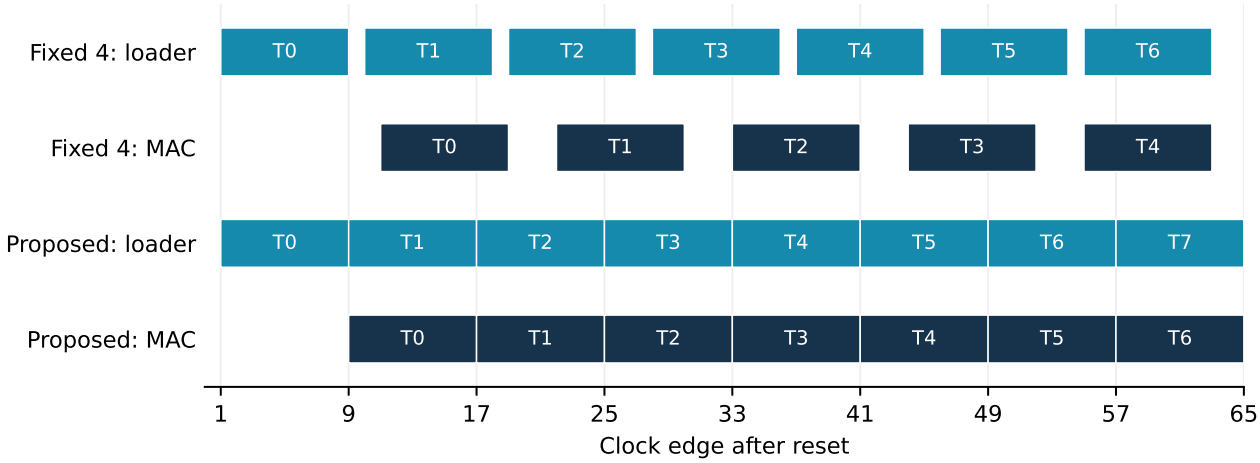


Figure 5: Loader and MAC execution for Fixed 4 and the proposed controller under the balanced workload. Each interval corresponds to an eight-clock tile operation. The proposed handoffs eliminate the three-clock MAC gap and the one-clock loader handoff gap.

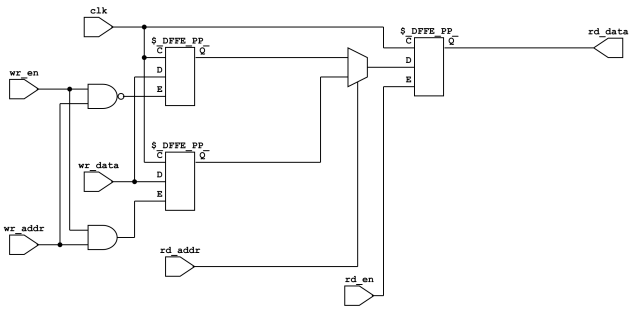


Figure 6: Gate-level implementation of a two-word, one-bit synchronous RAM. Two enabled flip-flops store the data; a multiplexer and output flip-flop provide the registered read path.

overhead.

Under variable latency, the proposed controller requires 925 cycles compared with 983 for Fixed 4, a 5.90% reduction, at an average allocation limit of 3.069. In the phase workload, it requires 1808 cycles compared with 1991 for Fixed 4, a 9.19% reduction, at an average limit of 2.619. The complete controller therefore improves execution time while reducing average allocation eligibility across these workloads.

5.2 Fixed and Variable Buffering

The buffering experiments use RH control to isolate bank capacity and allocation policy. At constant 8/8 latency, ping-pong reduces 32-tile execution from 640 to 361 cycles (1.773× versus single), but four banks offer no further improvement. With the deterministic variable profile, ping-pong takes 539 and fixed four-bank takes 497 cycles, a 1.0845× speedup over ping-pong. Its compute-idle count falls from 195 to 153 clocks. The same fixed-four result appears with static cyclic mapping, so this specific gain reflects additional independent tile slots rather than a measured benefit from dynamic arbitration at equal bank count.

The bank-count sweep in Fig. 8 for variable base 8/8 gives 539 cycles at two banks, 497 at three, and 497 at four, six, and eight. Thus the first additional slot matters for this pattern, while further slots saturate. This is not a general optimum bank count: there is still one transfer interface and one MAC, and the latency sequences are specific to the synthetic workload.

5.3 Runtime Adaptation and Phase Changes

Table 8 compares the proposed controller with ping-pong and Fixed 4. In the stable workloads, the proposed design averages 2.025 and 2.061 eligible banks and reduces idle time from 201 to ten clocks. Under variable latency, it requires 925 cycles at 74.81% utilization, compared with 983 cycles at 70.40% for Fixed 4.

For the 96-tile phase workload, the proposed controller requires 1808 cycles compared with 1991 for Fixed 4 (Table 9). Its 272 MAC-idle clocks are 183 fewer than Fixed 4's 455. All configurations perform 1536 MAC-active clocks, so the cycle difference corresponds directly to recovered overlap.

The proposed controller's trajectory in Fig. 9 starts at two eligible banks, briefly reaches three at execution cycle 32, and returns to two at 64. It rises to three at 672 and four at 704 as transfer latency becomes irregular, then falls to three at 1216 and two at 1248. Mode occupancy is 1200 clocks at limit two, 96 at three, and 512 at four, totaling 1808 clocks and an average limit of 2.619. The initial excursion indicates sensitivity of the starvation threshold to pipeline fill.

Table 6: Execution cycles for Fixed 4 and the proposed controller under identical per-tile service targets. Reduction is relative to Fixed 4; the average allocation limit refers to the proposed controller.

Workload	Fixed 4	Proposed	Reduction	Avg. limit
Compute-bound, 64 tiles	1481	1290	12.90%	2.025
Balanced, 64 tiles	713	522	26.79%	2.061
Memory-bound, 64 tiles	2123	2058	3.06%	3.953
Variable, 64 tiles	983	925	5.90%	3.069
Phase change, 96 tiles	1991	1808	9.19%	2.619
Minimum targets, 64 tiles	581	390	32.87%	2.000
Larger tile, 16 tiles	429	403	6.06%	2.079
One tile	20	18	10.00%	2.000

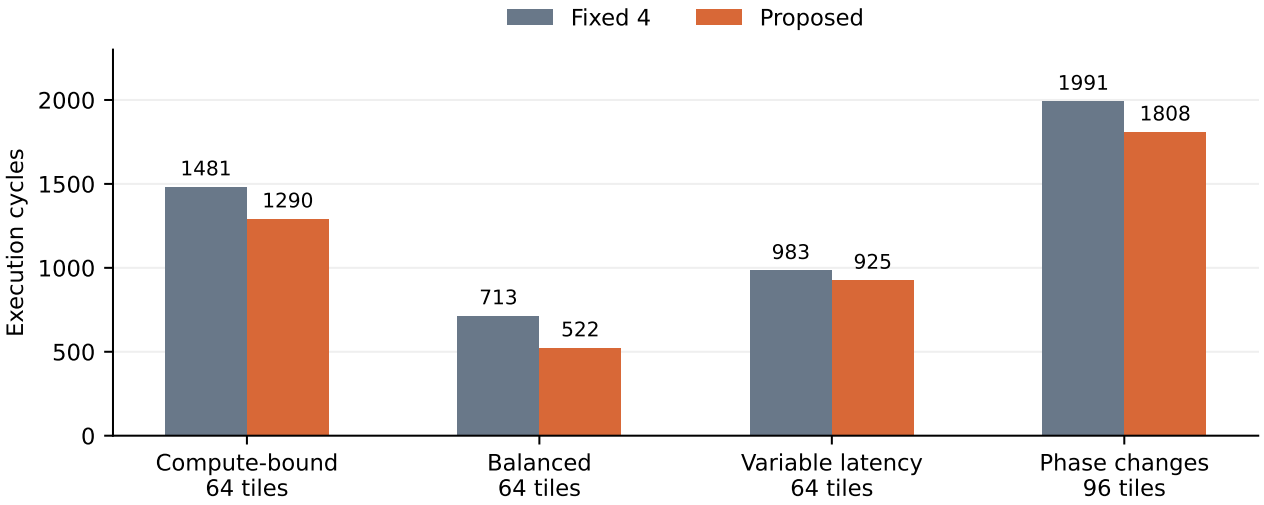


Figure 7: End-to-end execution cycles for Fixed 4 and the proposed controller across the four principal workloads.

Table 7: Execution cycles with RH control, 32 tiles, four pairs per tile, and base transfer/compute targets of 8/8.

Latency	Single	Ping-pong	Avail. 4	Cyclic 4
Fixed	640	361	361	361
Variable	922	539	497	497

Table 8: Baseline and proposed configurations for 64 tiles. Idle denotes MAC-idle clocks; $\bar{B}$ is the average allocation limit.

Configuration	Cycles	Idle	Util. (%)	$\bar{B}$
Compute-bound				
Ping-pong	1481	201	86.43	2.000
Fixed 4	1481	201	86.43	4.000
Proposed	**1290**	**10**	**99.22**	**2.025**
Balanced				
Ping-pong	713	201	71.81	2.000
Fixed 4	713	201	71.81	4.000
Proposed	**522**	**10**	**98.08**	**2.061**
Variable latency				
Ping-pong	1072	380	64.55	2.000
Fixed 4	983	291	70.40	4.000
Proposed	**925**	**233**	**74.81**	**3.069**

5.4 Equal Storage Capacity

Two eight-word banks and four four-word banks each instantiate 16 RAM words. Table 10 reports the variable and phase workloads at this storage capacity. Their cycle counts match those in the four-word-per-bank experiments because each bank holds at most one tile, irrespective of its addressable depth. Equal word count therefore does not imply equal independent tile capacity or decoder structure.

6 Discussion, Limitations, and Future Work

6.1 Buffer Capacity and Handoff Overhead

The buffering results distinguish overlap from capacity. Two-bank ping-pong overlaps loading and computation, while a third independent tile slot absorbs the transfer

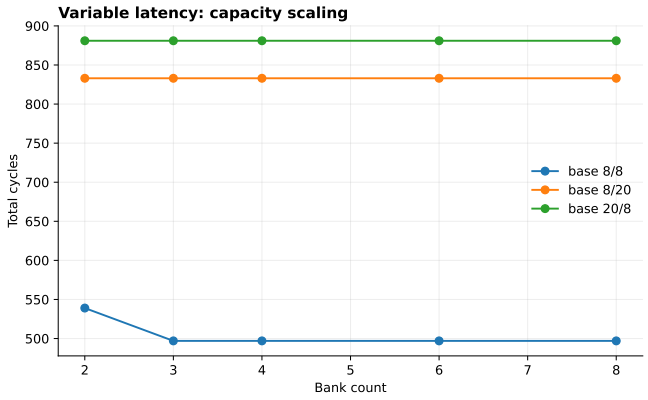


Figure 8: Bank-count sweep under the deterministic variable profile. Only the base 8/8 case benefits from moving from two to three banks in this sweep; additional banks do not reduce measured cycles.

Table 9: Baseline and proposed configurations for the 96-tile phase workload. All perform 1536 MAC-active clocks and 384 MAC operations.

Configuration	Cycles	Idle	Util. (%)	$\bar{B}$
Single	2950	1414	52.07	1.000
Ping-pong	2054	518	74.78	2.000
Fixed 4	1991	455	77.15	4.000
Proposed	**1808**	**272**	**84.96**	**2.619**

bursts in the variable 8/8 workload. At four banks, availability and static cyclic scheduling produce identical cycle counts in the matched RH experiments. The observed improvement over ping-pong is therefore attributable to independent tile capacity within the evaluated sequential tile stream.

The proposed design addresses control overhead and bank eligibility together. Completion-aware issue removes idle handoff clocks, while the allocation limit adapts to observed compute starvation. During the phase workload, eligibility increases to four banks and subsequently returns to two. The initial two-to-three excursion indicates sensitivity to pipeline fill, and the observation window introduces a finite response delay. Despite this delay, the complete controller reduces execution cycles relative to Fixed 4 under every workload in Table 6.

6.2 Model Scope and Limitations

The RAM model is behavioral RTL rather than a characterized SRAM macro. Four physical banks remain present even when the limit is two, so the experiment does not establish dynamic or leakage power savings, nor an area reduction. The model omits a DRAM protocol, a full CNN dataflow, cache hierarchy, physical timing, placement, and clock-frequency effects. A single loader also caps memory-bound throughput independently of bank count. The synthetic, deterministic latency profiles are useful controlled stresses but do not represent a measured application distribution. Bank depth and tile-slot count are entangled in the equal-word experiment.

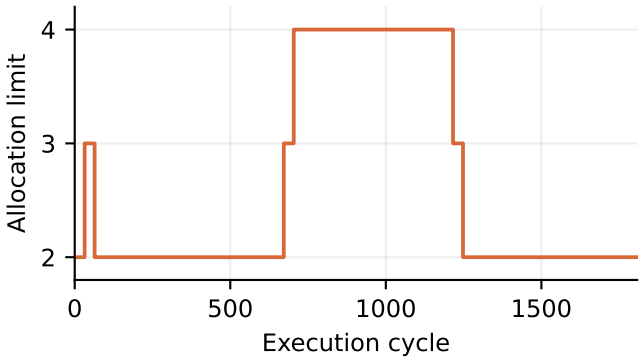


Figure 9: Runtime bank eligibility during the three-phase workload under the proposed controller.

Table 10: Execution cycles at equal storage capacity: two eight-word banks or four four-word banks, with four pairs per tile.

Case	Ping-pong	Fixed 4	Proposed
Variable, 64 tiles	1072	983	925
Phase, 96 tiles	2054	1991	1808

Cycle reduction assumes the same clock period. The look-ahead arbiter adds a completion-to-issue combinational path, so placement and timing are needed to establish an elapsed-time improvement. Generic synthesis cell counts provide a structural check, not a physical cost measurement. The comparison evaluates the combined controller against the specified Fixed 4 implementation. It does not establish that restricting bank eligibility alone reduces execution cycles.

Future evaluation will require application layer traces, sensitivity analysis of observation windows and thresholds, and characterization of startup adaptation. Multiple transfer channels would test the policy beyond the single-loader throughput limit. Technology mapping, SRAM macro integration, timing closure, and energy measurements are required to assess the physical cost of eligibility control and completion-aware issue.

7 Conclusion

Completion-aware handoffs reduce scratchpad execution cycles with the same four physical banks, one loader, one MAC, and unchanged tile service times. Combined with runtime eligibility, they reduce cycles by 26.79% in the balanced case, 5.90% under variable latency, and 9.19% under phase changes relative to Fixed 4. The corresponding average allocation limits are 2.061, 3.069, and 2.619 banks. These results characterize the complete proposed controller, which combines completion-aware handoffs with runtime bank eligibility. Physical timing and energy characterization remain necessary to evaluate

implementation efficiency.

8 Artifact Availability

The reproducibility package comprises parameterized Verilog RTL, a self-checking testbench, deterministic workload generators, simulator logs, CSV summaries, VCD traces, and logic-synthesis scripts. Each performance record is accepted only after arithmetic and protocol checks pass. The evaluation dataset includes per-tile acceptance and completion events, configuration parameters, and source hashes. Plotting scripts derive the reported figures directly from these records. The package README specifies tool setup and commands for simulation, independent analysis, synthesis, and figure generation.

References

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